

Partition-Tolerant Federation Across Simulated ISP Islands

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Abstract

National disconnection does not necessarily remove all domestic IP reachability: access networks may remain as isolated “islands” even when international routes disappear. We investigate whether a signed, server-federated forum can keep accepting and carrying public posts across such residual paths. The prototype admits byte-canonical protobuf envelopes through a 19-stage validation pipeline, returns signed transparency receipts, initiates synchronization from either side, and permits an explicitly trusted dual-homed node to relay between otherwise disjoint islands. In a four-node container deployment, after removing every inter-island route, certificate, community, and post objects crossed a bridge in 0.5, 1.2, and 2.1 seconds. After repairing serial head-of-line blocking, the post case fell from 407.6 seconds to 2.1 seconds. In a separate eight-node in-process chain, 200/200 envelopes reached hop 7 (median 4.125 seconds; p99 6.115 seconds). TypeScript, Rust, and Python implementations agreed on all 16 canonical vectors. A 3,000-request malformed-envelope run sustained 730 requests/s, but 90% of requests were rate-limited; the defensible result is zero invalid-object writes, not denial-of-service resistance. Against 80 measured ActivityPub activities, the proposed encoding used 4.4–6.9× fewer bytes than gzipped JSON-LD (19.5–25.9× fewer than raw JSON-LD, with caveats). These controlled, single-host results establish a reproducible mechanism for partition-tolerant federation under surviving domestic IP paths; they do not establish operation in a total communications blackout or resistance to a coercive network operator.

CCS Concepts

• **Networks** → **Network reliability**; • **Security and privacy** → *Distributed systems security*; • **Information systems** → *Social networks*.

Keywords

network partitions, federated systems, signed objects, transparency logs, canonical encoding, internet shutdowns

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1 Introduction

Shutdown measurements increasingly show a spectrum rather than a binary state: throttling, protocol interference, destination blocking, and loss of international routes can leave unequal fragments of connectivity [3, 18]. Baltra et al. call such mutually reachable fragments *connectivity islands* [1]. This paper asks a deliberately narrow question: if some domestic IP routes survive,

can independently operated forum servers keep accepting authenticated objects and synchronize them across the remaining graph, including through a locally trusted relay?

Existing circumvention and blackout systems answer adjacent questions. Proxies and refraction networking reconnect users to infrastructure outside the filtered region [5, 8, 26]. Device-to-device systems disseminate messages without IP infrastructure [12, 13, 16, 17]. Federated social protocols assume routable server addresses, while signed-log systems provide integrity without an operator-oriented admission and receipt workflow [4, 7, 22, 23, 25]. The unfilled design point is therefore neither global circumvention nor infrastructure-free delay-tolerant networking. It is *server federation inside partial reachability*, with verifiable acceptance at each operator boundary.

We present and evaluate a working implementation with four contributions:

- a single byte-preserving admission path for local and federated signed objects, with no persistent writes in validation stages 1–12;
- caller-initiated delivery, streaming, and backfill, allowing a server that cannot accept inbound connections to participate by dialing a reachable peer;
- policy-constrained bridging by a normal validating server that is explicitly trusted by each side and cannot relay an object back over its incoming uplink; and
- reproducible evidence from canonical vectors, unit and integration suites, a route-cut container experiment, an eight-node chain, and a measured encoding comparison.

Our claim is correspondingly limited. The system is *partition tolerant under surviving IP paths*. It cannot manufacture a route through a physical or radio blackout, hide traffic from deep packet inspection, or prevent an operator from refusing valid posts.

2 Background and Research Gap

2.1 Existing solution families

Circumvention systems conceal or redirect a user’s connection to an external service. Telex places anti-censorship logic in cooperating ISP routers [26]; domain fronting and Snowflake exploit alternative rendezvous or proxy surfaces [5, 8]. These are valuable when an external path exists, but do not make domestic servers converge after that path is removed.

Blackout messaging removes the server assumption. Twilight stores and forwards tweets over opportunistic Bluetooth contacts [12]; Rangzen and Anix combine encounters with social trust or endorsement [13, 16]; Moby targets anonymous messaging under network blackouts [17]. Dolphin instead tunnels through cellular voice at low bit rate [20]. These systems trade latency, energy, or bandwidth for independence from domestic IP infrastructure. They do not define a federated public-forum boundary between operators that still run reachable servers.

Federation and signed replication provide reusable pieces. ActivityPub defines inbox-based social federation [25]; Matrix and

AT Protocol provide state or repository federation [4, 23]; Nostr and Secure Scuttlebutt sign replicated content [7, 22]. FETHR explored decentralized micropublishing at Internet scale [19], while Publius replicated static censorship-resistant documents [24]. These systems generally presume routable destinations, or lack the combination of raw-byte admission, operator receipts, and an explicit two-island relay policy studied here.

Accountability is not availability. Certificate Transparency (CT) makes an append-only log auditable with signed tree heads and consistency proofs [14, 15]; PeerReview and CoSi show stronger paths to distributed accountability [10, 21]. Borrowing CT-style receipts can prove that a server acknowledged an object. It cannot prove that the server admitted every valid object, forwarded promptly, or resisted coercion.

2.2 Gap and design objective

Prior work separates into three substrates: globally routed circumvention, infrastructure-free encounter networks, and normally routed federation. Hasan identifies the wider challenge of scaling blackout circumvention while preserving operator safety [11]; Baltra et al. provide the vocabulary for reasoning about partial reachability [1]. We target the gap between them: IP works *within* or *between some* islands, servers remain online, but DNS, inbound reachability, or international routing may be unreliable. The objective is to accept an authored object once, preserve its signed bytes, and make every reachable replica eventually process the same object under the same rules.

3 System Design

3.1 Object and admission model

An author creates a protobuf envelope containing a domain tag, object type, payload, public key, nonce, and expiry. Its identifier is the SHA-256 digest of the specified canonical bytes, and an Ed25519 signature covers those bytes [2]. Protobuf does not promise canonical serialization [9]; the profile therefore fixes field order, rejects unknown fields, uses minimal variants, and requires decode–re-encode byte equality. The receiver never “re-pairs” attacker-controlled bytes before signature verification.

Local and federated objects enter the same 19-stage pipeline. Stages 1–12 enforce raw size, parse and canonicity, domain separation, identifiers, signature, expiry, nonce/replay, and semantic authorization without persistent writes. Later stages apply origin admission or receiver-side peer quotas, commit the exact raw envelope and derived indexes transactionally, append its hash to a Merkle log, and enqueue fan-out only after commit. Exact retries are idempotent. Local admission may require a priced proof; federated reception is quota-bounded because a malicious origin can omit its own price. We therefore do not claim that origin pricing protects other operators.

On acceptance, the server returns a signed receipt containing the object identifier, leaf position, signed tree head, and inclusion proof. A client can bind acknowledgment to the submitted bytes. Receipts make accepted-history equivocation detectable when peers compare sufficient evidence, but they do not prevent refusal or selective delay.

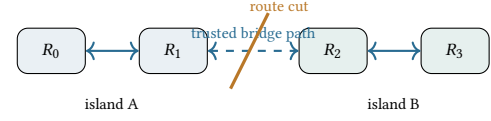


Figure 1: Four-node virtual topology. The dashed path is application-level storage and relay, not transparent IP forwarding. All containers run on one physical host.

3.2 Partition-aware federation

Three caller-initiated RPCs cover the synchronization cases: DELIVER sends a batch, STREAMACTIVITIES resumes from a durable cursor, and BACKFILL retrieves missing objects. Outbox queues are grouped by source uplink, destination peer, and traffic plane, then drained concurrently. Each attempt has a scheduler deadline, exponential backoff, and an idempotency key. The underlying library call is not forcibly cancelled at the deadline, a limitation discussed in Section 7. Source-address binding and a resolver keyed by source and destination allow one process to select among simulated ISP uplinks.

Figure 1 shows the deployed topology. R_0 and R_1 share island A; R_2 and R_3 share island B. R_1 is dual-homed. After the inter-island route is cut, ordinary nodes cannot dial across islands. Each side explicitly lists R_1 as a trusted bridge. The bridge is not a packet forwarder: it verifies and stores each envelope as a normal server, applies a token bucket per uplink pair and object class, and never emits an object over the uplink on which it arrived. Bulk backfill receives half the interactive rate. These rules bound relay volume and loops, but do not make the bridge covert or safe from coercion.

Peers gossip signed tree heads (STHs). Same-size/different-root and a smaller new head at a later trusted time are strong conflicts. An older timestamp is ignored as stale. A larger tree is recorded as *pending* rather than declared consistent: growth requires a CT consistency proof. Thus the current mechanism provides stale-head suppression and some conflict detection, not complete fork detection for growing trees.

3.3 Replication and recovery invariants

The implementation enforces four invariants at adapter boundaries. First, federation transports an opaque byte string into ingress. A generated RPC decoder must not deserialize and then serialize an envelope, because doing so could silently normalize a non-canonical attack into an accepted object. Second, peer trust changes volume and priority policy, never validity: even an explicitly configured bridge must pass signature, identifier, expiry, replay, and authorization checks. Third, deduplication is a database uniqueness constraint on content identifier and direction rather than a read-before-write check. This matters when a live stream and a backfill concurrently deliver the same object. Fourth, fan-out excludes the origin peer. Content deduplication would eventually stop a loop, but origin exclusion avoids paying a redundant round trip for every accepted object.

Identifiers for projected entities are derived only from signed envelope fields. This rule was added after an early projector used

the receiving node’s key when constructing a community identifier. A single-node test could not reveal the error; two nodes produced permanently different projections from the same signed object. The corrected implementation asks whether any node with only the signed bytes will derive the same identifier. Dependency objects are admitted before their dependents, and missing dependencies remain retryable rather than being partially projected.

Durable cursors make recovery independent of a continuously open connection. A caller presents its last accepted cursor, the peer streams later activities, and exact retries are absorbed by the unique index. Backfill uses the same path for an explicit missing range. Consequently, a node advertising no inbound endpoint can still send its own outbox and retrieve the peer’s history over connections that it initiates. This supports asymmetric reachability such as carrier-grade NAT, but requires at least one side to know a reachable peer address. It is not peer discovery.

Strict canonicity also constrains evolution. An old implementation rejects unknown fields instead of forwarding data it cannot authenticate under its own profile. Compatible schema changes therefore need a versioned domain or envelope profile and cross-language vectors for both versions. We accept this upgrade cost to keep the signed byte string unambiguous.

4 Methodology

We evaluate five questions: (RQ1) do independent encoders agree; (RQ2) do objects cross an enforced route cut; (RQ3) does propagation remain complete over multiple hops; (RQ4) do malformed requests become storage writes; and (RQ5) what is the wire-size difference from a deployed federation format?

Correctness suites. A shared corpus contains 16 positive and adversarial byte vectors. It is executed independently by TypeScript, Rust, and Python implementations. The backend suite exercises the ingress, outbox, federation, and ISP abstractions. At revision time, 85 focused tests passed (29 ingress, 3 outbox, 31 federation, and 22 ISP tests), in addition to the vector gate. These are in-process checks and are not counted as deployment measurements.

Four-node route-cut deployment. Four application containers, four databases, and four caches are attached to three private Docker networks. The gate records source/destination reachability from kernel TCP state, removes the only ordinary inter-island path, proves that direct island-A to island-B probes fail, submits a dependency chain (issuer certificate, community, post), and polls the destination for each exact content identifier. The bridge has fixed addresses on both island networks. Reported times are end-to-end wall-clock observations from this gate, including application polling. Because all containers share one physical host, the experiment validates routing and software state transitions, not geographic latency, independent failure domains, or ISP behavior.

Eight-node chain. An in-process harness links eight servers in a line. It injects 200 valid envelopes at hop 0, waits for all to reach hop 7, and records per-object completion latency. This isolates the federation scheduler but omits container networking and durable external services.

Invalid-envelope run. A client sends 3,000 over-sized or non-canonical requests at concurrency 16 to one node. We measure

completion rate, rejection classes, signature-verification entries, and persistent object count. The node’s normal limiter remains enabled; hence throughput is descriptive of the configured gate, not a saturation benchmark.

Encoding comparison. We collected 80 activities emitted by two stock ActivityPub implementations, retained the delivered JSON-LD contexts, and compared delivered and gzip-compressed bytes with equivalent signed protobuf envelopes. We subtract authored text to estimate overhead but do not include HTTP framing, ActivityPub HTTP signatures, TLS records, follower fan-out, or compression dictionaries. The comparison addresses representation size, not total protocol cost.

Reproducibility boundary. The artifact contains the compose topology, route gate, canonical fixtures, and deterministic test entry points; generated bindings are checked for drift. The focused suites and vector gate were rerun for this revision. Route-cut numbers come from the deployment gate’s preserved output. Random seeds, host load, repeated-run distributions, and confidence intervals were not preserved for those observations; they should be added before drawing performance-generalization claims.

5 Results

5.1 Correctness and convergence

RQ1. All three independent implementations produced identical output for 16/16 canonical vectors and rejected every negative vector as specified. This is evidence that the wire profile is implementable across languages; sixteen fixtures are not a proof that all protobuf edge cases are covered.

The focused suites also exercise failures that the happy path can hide: corrupted signatures, non-canonical encodings, identifier disagreement, replay, missing authorization dependencies, concurrent duplicate delivery, a receipt under the wrong key, an inclusion proof for different content, stale STHs, same-size/different-root conflicts, and shrinking trees. Growing STHs are expected to enter the pending state, reflecting the missing consistency proof rather than being reported as safe. The 85-test count is useful regression evidence, but it combines unit and in-process integration tests and is not an independent security audit.

RQ2. The deployment gate completed 19/19 assertions. With direct inter-island reachability removed, the destination observed the certificate after 0.5 seconds, the dependent community after 1.2 seconds, and the post after 2.1 seconds. A prior implementation drained destinations serially and allowed one blackholed peer to block every following peer; the same post path then took 407.6 seconds. Grouping queues per peer/uplink and draining them concurrently reduced this observation by 99.5%. The result explains the improvement: it removes unrelated head-of-line wait, rather than accelerating cryptography or transport.

RQ3. All 200 envelopes reached hop 7. End-to-end median was 4.125 seconds and p99 was 6.115 seconds. Figure 2 shows the monotonic hop cost. The modest p99–median gap is consistent with bounded parallel draining in this low-contention harness; the experiment does not model WAN jitter or adversarial valid traffic.

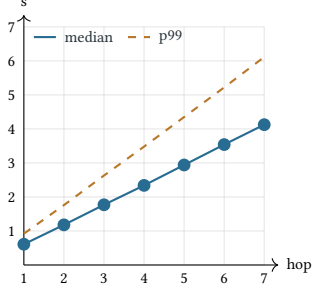


Figure 2: Eight-node in-process chain, 200/200 envelopes delivered. Intermediate-hop values are recorded observations; hop 7 is the reported end-to-end result.

Table 1: Representation bytes over 80 measured ActivityPub activities. Ratios compare ActivityPub with the proposed envelope after subtracting authored text.

	Post	Follow	Delete
Proposed envelope	206	142	250
ActivityPub, gzip	1,074	786	1,400
Gzip ratio	5.2×	5.5×	5.6×
Observed range	4.4–6.9×		
ActivityPub, raw	4,216	2,786	6,909
Raw overhead ratio	19.5×	19.5×	25.9×

5.2 Admission and representation cost

RQ4. The invalid run completed at 730 requests/s and produced zero object, index, log, or outbox writes. However, 2,700/3,000 requests (90%) were stopped by rate limiting; only about 9% entered signature verification. The sound conclusion is that the tested malformed traffic did not amplify into invalid-envelope storage writes. It is not evidence of denial-of-service resistance: valid signed floods, distributed sources, parsing allocation, and database contention remain open.

RQ5. Table 1 leads with the compression-aware comparison. For matched post, follow, and delete objects, gzipped ActivityPub was 4.4–6.9× larger. Raw JSON-LD overhead was 19.5–25.9× larger, but that ratio advantages a binary format by ignoring routine HTTP compression. The gzip result is the more operationally relevant observation.

5.3 Qualitative and quantitative comparison

Table 2 places the results beside systems with different substrates and threat models. The values are intentionally not normalized: “delivery” means server-to-server convergence here, encounters in Anix/Rangzen/Twilight, modeled visibility in GhostPost, and a trace-driven flood in Moby. Sample size and evaluation method are therefore shown in the same cell.

The comparison exposes complementarity rather than dominance. Encounter systems operate when IP is absent but converge over hours and consume scarce handset energy. The proposed server path converges in seconds in a virtual topology but requires powered, mutually reachable infrastructure. Dolphin can

cross a packet-data shutdown but its 64-bit/s channel is unsuitable for routine federation. Normally routed federation and static replication scale to larger observed corpora, yet their cited evaluations do not test the route-cut and dual-homed relay case. Our experiment fills that narrow gap; it is far smaller than the trace and simulation studies in Table 2.

6 Discussion

Why did the bridge result improve? The 407.6-second failure was a scheduling artifact: one unreachable destination monopolized a serial drain. Per-peer concurrent queues make progress on a reachable bridge independent of that destination, producing the 2.1-second result. Deadlines alone would cap, rather than remove, cross-peer interference; unrestricted concurrency would improve an idle test yet allow a valid flood to consume memory and sockets. The chosen bounded concurrency is a middle path, although the uncanceled underlying call still deserves repair.

What if we had chosen device-to-device networking? Bluetooth or Wi-Fi Direct would extend operation to a total IP blackout and remove the dual-homed-server dependency. Prior results suggest a different operating regime: encounter-based convergence in hours, discovery overhead, handset battery cost, and difficult background execution. A hybrid design could carry the same signed envelope over opportunistic links, but peer discovery, privacy, duplicate suppression, and mobile energy would become primary research questions. Our seconds-scale server results cannot predict that hybrid’s behavior.

What if we had reused ActivityPub JSON-LD? Interoperability and an existing application ecosystem would improve. The measured raw representation would be substantially larger, though gzip narrows the difference to 4.4–6.9×. More importantly, signing ActivityPub across JSON-LD processing requires an agreed canonicalization and authentication profile. Conversely, selecting a custom canonical protobuf reduces bytes and makes the verified byte string explicit, but creates a new interoperability burden and a risk that untested codecs disagree. The 16-vector, three-language gate reduces rather than eliminates that risk.

What if receivers normalized incoming bytes? Decode–normalize–verify would make benign codec variation easier to accept and would ease rolling upgrades. It would also make the receiver, rather than the author, choose the signed representation. Two libraries could accept different byte strings for the same logical message, complicating content identifiers, replay detection, and receipt evidence. Raw-byte preservation keeps those properties simple, while versioned profiles carry the compatibility cost. An alternative worth evaluating is a formally specified deterministic protobuf subset with differential fuzzing substantially larger than the present 16-vector corpus.

Global consensus ordering. A Byzantine consensus group or witness-cosigned log could strengthen global non-equivocation [21]. During a partition, however, quorum placement would decide which island can write, undermining local availability. The current per-operator logs preserve independent admission and later evidence exchange. The price is weaker accountability: a receipt proves acknowledged inclusion only, and growing STHs remain pending until consistency proofs are implemented.

Table 2: Comparison with representative prior systems. Values retain each paper’s own denominator and workload; they are context, not a cross-system ranking. “IP federation” means an evaluated server-to-server path under partial reachability.

System	Substrate	Public/authored integrity	Reported quantitative evidence	Partial-IP federation	Principal distinction
This work	Surviving IP; trusted dual-uplink relay	Yes; Ed25519 envelope + receipt	4-node route cut: 0.5/1.2/2.1 s; 8-node chain: 200/200, hop-7 p50 4.125 s, p99 6.115 s	Yes, four-node virtual topology	Operator-oriented signed admission and resumption
Anix [13]	Bluetooth/Wi-Fi Direct	Public anonymity + endorsement	600 simulated users; >90% dissemination in about 23 h; discovery 1.8 s, exchange 11.58 s	No	Infrastructure-free opportunistic spread
Rangzen [16]	Phone encounters	Anonymous social-trust dissemination	400-node main simulation, 40-run means; >80% in 24–48 h; encounter 6.61 s	No	Friend-to-friend trust under blackout
Moby [17]	Mobile opportunistic network	Private anonymous messaging	268,596-user trace basis; highlighted flood 13.96% vs. 1.15% epidemic delivery	No	Stronger privacy/forward secrecy objective
Twilight [12]	Bluetooth store-and-forward	Authenticated disaster tweets	5 users, 915 tweets; almost all delivered in about 2 h; 41–68 h battery	No	Early deployed disruption-mode client
Dolphin [20]	Cellular voice to gateway	Encrypted low-rate tunnel	64 bit/s, <2% error; secure setup about 60 s; 500-character email about 160 s, 30 trials	No	Works without packet data, at low throughput
GhostPost [6]	Browser caches/social graph	Restores already observed posts	1M simulated users; about 1.5% adoption preserves a majority of modeled 30-min views	No	Restoration after platform deletion
Secure Scuttlebutt [22]	Peer replication	Signed append-only author logs	Operational community reported above 10,000; no controlled partition-latency result	No server path evaluated	Closest signed offline log lineage
FETHR [19]	Routable HTTP/callback	Decentralized micropublishing	4,917,042-message/472,735-user trace; about 120k items processed by prototype	Normally routed only	Large trace, but no route-cut metric
Publius [24]	Reachable replicated web servers	Threshold-based static publication	Prototype of about 1,500 lines; analytical/functional evaluation	Normally routed only	Static-document availability

What if bridges were automatic? Discovering and accepting any dual-homed node would improve reachability but turn it into an attractive Sybil, eclipse, and traffic-amplification surface. Explicit bilateral trust limits that surface and makes operator responsibility visible, at the cost of manual configuration and a conspicuous chokepoint. Multiple disjoint bridges and signed policy distribution would reduce single-relay dependence, but were not evaluated.

Interpreting the measured boundaries. The multi-hop curve grows roughly linearly because each hop waits for local validation, storage, and the next bounded drain. That behavior is appropriate for a short server path and less attractive for a long overlay. The zero-write invalid run follows from placing cheap byte and canonical checks before state changes, but the 90% limiter share means it mostly demonstrates configured admission ordering. Finally, the representation result is largest against uncompressed JSON-LD and smaller against gzip; compression-aware numbers should guide deployment expectations.

7 Limitations, Threats, and Ethics

External validity. The strongest limitation is scale and realism. The deployed topology has four nodes on one physical machine; the chain has eight in-process nodes; there is one measured route-cut run reported by the gate. Shared CPU, clock, storage, and kernel remove geographic latency and correlated infrastructure failures. No field shutdown, ISP cooperation, mobile client, human moderation workload, or multi-day reconnection was studied. The related-work quantities use different substrates and denominators and must not be read as a leaderboard.

Network adversary. A domestic operator can identify stable endpoints, apply IP or SNI whitelists, fingerprint gRPC/TLS traffic, perform traffic analysis, coerce a bridge, or isolate nodes into

an eclipse. Source binding does not conceal an endpoint. A total power, radio, and packet outage leaves no path. The design should therefore be described as partition tolerant, not censorship resistant. Operational exploration would require endpoint rotation, pluggable transports, bridge diversity, and explicit risk assessment for local operators.

Application adversary. Signatures authenticate keys, not truth or safety. A malicious author can submit abuse; a valid-key flood can pass the early cryptographic gate; a malicious origin can skip its own admission price; and receiver quotas may still permit costly work. Key theft, metadata leakage, moderation, revocation recovery, storage exhaustion, and coordinated Sybil behavior remain open. The scheduler deadline does not cancel the underlying RPC. The current STH rule suppresses stale heads and detects some conflicts, but cannot establish append-only growth without consistency proofs.

Accountability boundary. Operator-signed receipts are evidence of acknowledged acceptance, not proof of universal availability. A server can refuse to issue a receipt, selectively withhold an accepted object, show different histories to disconnected peers, or delay until evidence expires. Independent witnesses, consistency proofs, and eventually PeerReview- or CoSi-like mechanisms could strengthen detection, but cannot force service under coercion.

Ethics. The experiments used synthetic content and isolated private networks; they did not interact with production services or real users. A real deployment could expose bridge operators and users to legal or physical risk. Publication of a protocol should not be mistaken for advice to run a visible relay in a hostile jurisdiction. Threat modeling, informed consent, data minimization, and local legal and community review are prerequisites for field work.

8 Conclusion

This work demonstrates a narrow but useful point: when domestic IP fragments remain, server federation need not stop merely because normal inter-island routing does. A byte-canonical signed object, common validation path, caller-initiated synchronization, and explicitly trusted validating bridge carried a dependency chain across an enforced virtual route cut, while bounded per-peer draining removed a severe head-of-line delay. Cross-language vectors and an eight-node chain provide additional correctness evidence, and the comparison clarifies where this server-based design sits relative to encounter, voice-channel, and normally routed systems. The evidence is controlled and small-scale. The next steps are CT consistency proofs, cancellation-aware transport, multi-host and wide-area experiments, valid-traffic stress tests, multiple independent bridges, and a hybrid opportunistic transport for periods in which IP disappears entirely.

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